OS LAB 02

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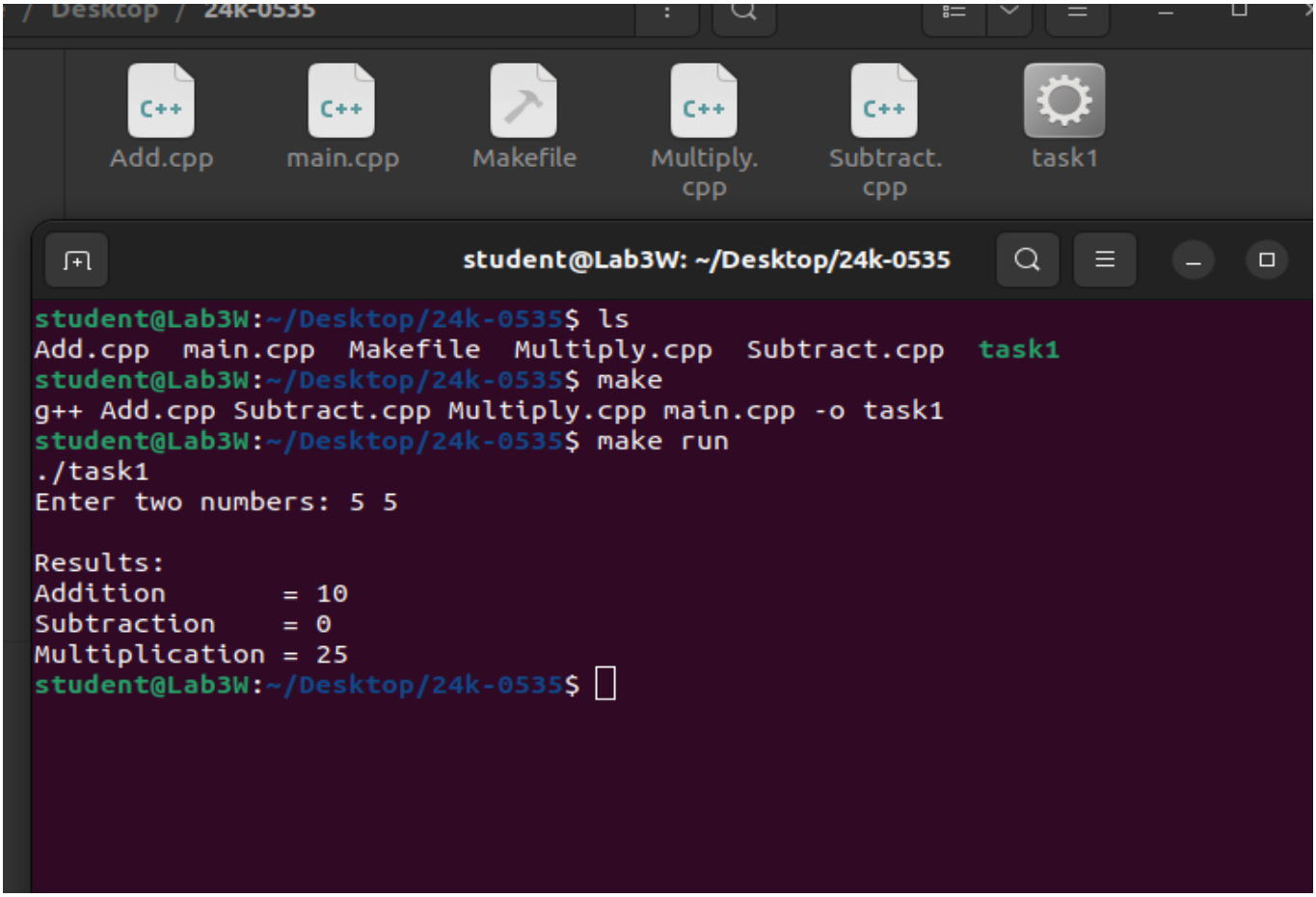
Roll No 24K-0535

Task 1 (Screenshots)



Add.cpp main.cpp Makefile Multiply.cpp Subtract.cpp

student@Lab3W: ~/Desktop/24k-0535

```
student@Lab3W:~/Desktop/24k-0535$ ls
Add.cpp  main.cpp  Makefile  Multiply.cpp  Subtract.cpp  task1
student@Lab3W:~/Desktop/24k-0535$ make
g++ Add.cpp Subtract.cpp Multiply.cpp main.cpp -o task1
student@Lab3W:~/Desktop/24k-0535$ make run
./task1
Enter two numbers: 5 5

Results:
Addition       = 10
Subtraction    = 0
Multiplication = 25
student@Lab3W:~/Desktop/24k-0535$ make clean
rm -f task1
student@Lab3W:~/Desktop/24k-0535$
```

Open ~/Desktop/24k-0535

Add.cpp

```cpp
#include <iostream>
using namespace std;

int add(int a, int b) {
    return a + b;
}

```

Open Subtract.cpp ~/Desktop/24k-0535

Add.cpp

```cpp
#include <iostream>
using namespace std;

int subtract(int a, int b) {
    return a - b;
}

```

Open

Multiply.cpp
~/Desktop/24k-0535

Add.cpp × Subtract.cpp

```cpp
#include <iostream>
using namespace std;

int multiply(int a, int b) {
    return a * b;
}

```

Open

main.cpp
~/Desktop/24k-0535

Add.cpp × Subtract.cpp × Multiply.cpp

```cpp
#include <iostream>
using namespace std;

// Function declarations
int add(int a, int b);
int subtract(int a, int b);
int multiply(int a, int b);

int main() {
    int a, b;

    cout << "Enter two numbers: ";
    cin >> a >> b;

    cout << "\nResults:\n";
    cout << "Addition       = " << add(a, b) << endl;
    cout << "Subtraction    = " << subtract(a, b) << endl;
    cout << "Multiplication = " << multiply(a, b) << endl;

    return 0;
}

```

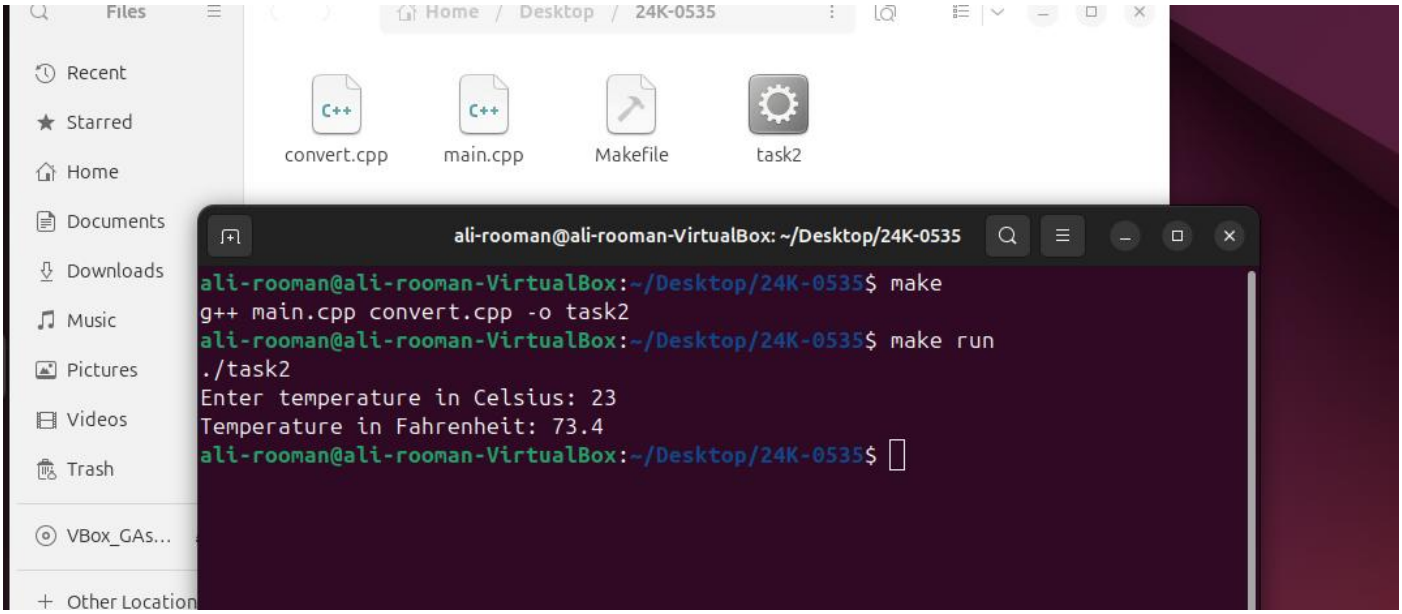
Open

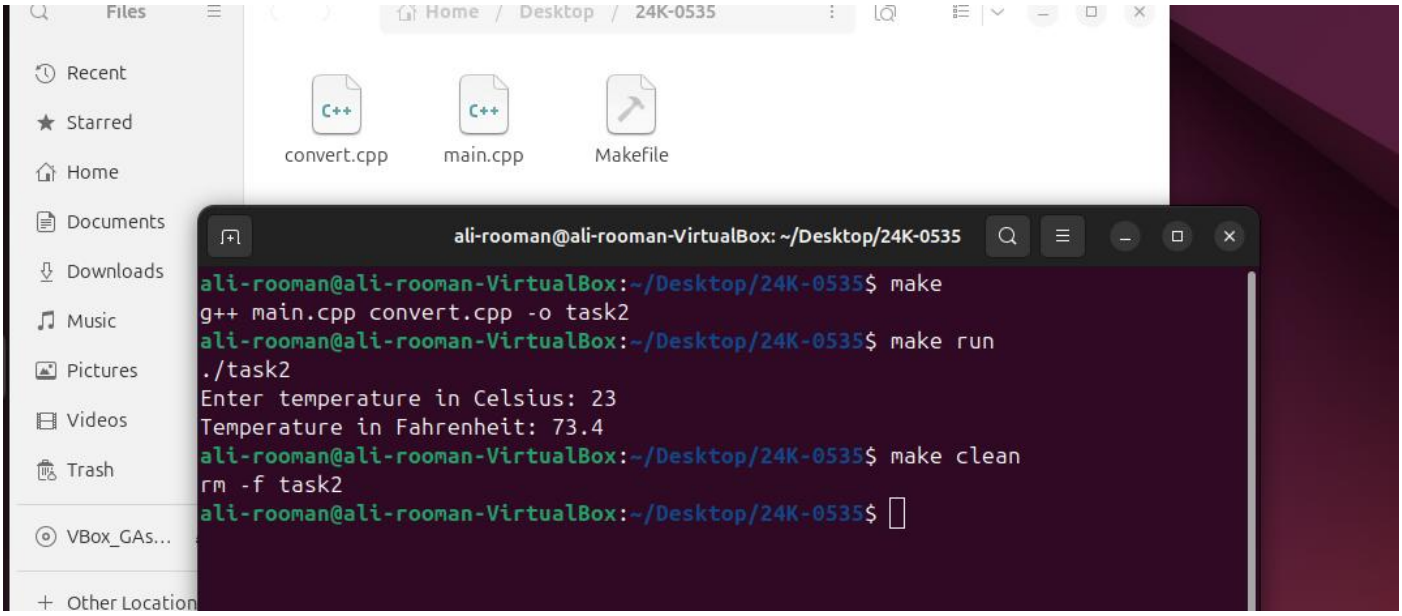
Makefile
~/Desktop/24k-0535

Add.cpp × Subtract.cpp × Multiply.cpp × ma

```makefile
task1: Add.cpp Subtract.cpp Multiply.cpp main.cpp
        g++ Add.cpp Subtract.cpp Multiply.cpp main.cpp -o task1
run: task1
        ./task1
clean:
        rm -f task1
```

Task 2 (Screenshots)





Open convert.cpp ~/Desktop/24K-0535

```cpp
double celsiusToFahrenheit(double celsius) {
    return (celsius * 9.0 / 5.0) + 32;
}
```

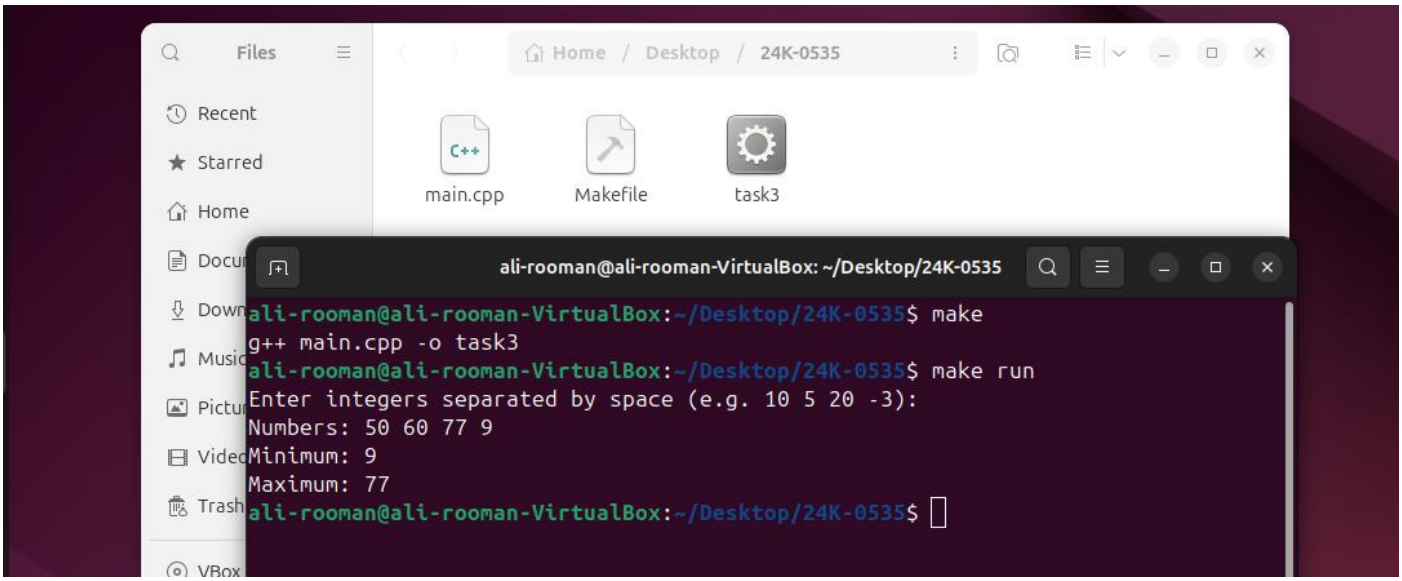
Open Makefile ~/Desktop/24K-0535

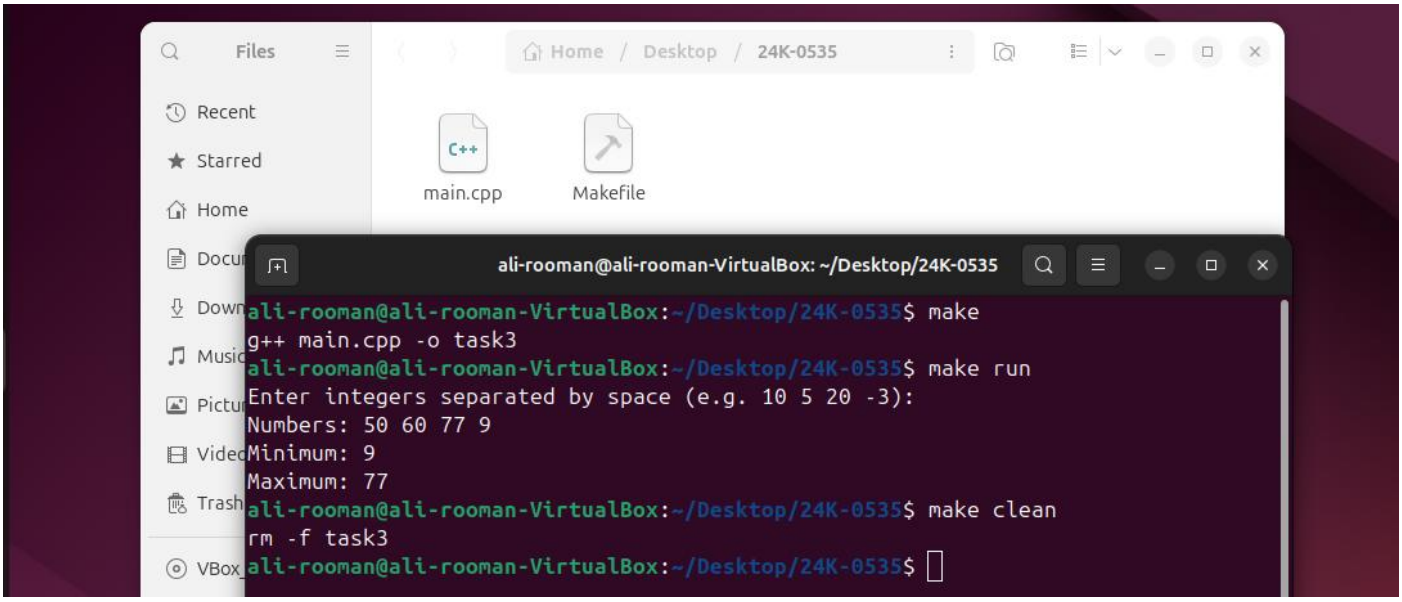
```makefile
task2: main.cpp convert.cpp
        g++ main.cpp convert.cpp -o task2

run: task2
        ./task2

clean:
        rm -f task2
```

Open main.cpp ~/Desktop/24K-0535

```cpp
#include <iostream>
using namespace std;


double celsiusToFahrenheit(double);

int main() {
    double celsius;

    cout << "Enter temperature in Celsius: ";
    cin >> celsius;

    double fahrenheit = celsiusToFahrenheit(celsius);

    cout << "Temperature in Fahrenheit: " << fahrenheit <<
endl;

    return 0;
}
```

Task 3 (Screenshots)





Open ~/Desktop/24K-0535

```
# Task 3 executable
TARGET = task3
SRC = main.cpp

# Compile the program
all: $(TARGET)

$(TARGET): $(SRC)
	g++ $(SRC) -o $(TARGET)

# Run interactively
run: $(TARGET)
	@echo "Enter integers separated by space (e.g. 10 5 20 -3):"
	@read -p "Numbers: " args; ./$(TARGET) $$args

# Clean compiled files
clean:
	rm -f $(TARGET)

.PHONY: all run clean
```

Open • main.cpp ~/Desktop/24K-0535

```
S#include <iostream>
using namespace std;

bool isNumber(string s) {
    if (s.length() == 0)
        return false;
    for (int i = 0; i < s.length(); i++) {
        if (i == 0 && s[i] == '-')
            continue;
        if (s[i] < '0' || s[i] > '9')
            return false;
    }
    return true;
}
int main(int argc, char* argv[]) {
    if (argc < 2) {
        cout << "Please provide integers as command line arguments.\n";
        return 1;
    }
    int arr[argc - 1];
    for (int i = 1; i < argc; i++) {
        string input = argv[i];
        if (!isNumber(input)) {
            cout << "Invalid input: " << input << endl;
```